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0800MAT201122101

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Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech S3 (R,S) / S1 (PT) (S, FE) Examination December 2023 (2019 scheme)

Course Code: MAT201

Course Name: Partial Differential equations and Complex analysis

Duration: 3 Hours

Max. Marks: 100

PART A

Answer all questions. Each question carries 3 marks

Marks

1

Derive partial differential equation from $z = f(x + 2t) + g(x - 2t)$

(3)

2

Solve $xp + yq = z$.

(3)

3

Find the steady state temperature distribution in a rod of length 30 cm if the ends are kept at 10°C and 100°C .

(3)

4

A tightly stretched string of length 50 cm has its ends fastened at $x = 0$ and $x = 50$. The midpoint of the string is then taken to height h and then released from rest in that position. Find the initial position of the string.

(3)

5

Check whether the function $f(z) = \begin{cases} \frac{\partial u(z)}{\partial \bar{z}} & z \neq 0 \\ 0, & z = 0 \end{cases}$ is continuous at $z = 0$.

(3)

6

Is $f(z) = \bar{z}$ analytic. Justify your answer.

(3)

7

Evaluate $\int_0^{1+i} (x - iy)dz$ along the straight line path from 0 to $1+i$

(3)

8

Evaluate $\int_C \frac{dz}{(z^2+1)(z-1)}$ when C is $|z| = 0.5$

(3)

9

Find the residue of $\sec z$ at $z = 0$

(3)

10

State Cauchy's integral formula and Cauchy's residue theorem

(3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11

(a) Form the partial differential equation by eliminating the arbitrary function f from the relation $xyz = f(x + y + z)$.

(7)

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- (b) Solve $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = u$ with the initial condition $u(x, 0) = e^{2x}$ by applying method of separation of variables. (7)

- 12 (a) Solve $\frac{\partial u}{\partial x \partial y} = \cos(x) \cos(y)$ given that $\frac{\partial u}{\partial y} = e^{-x}$ when $x = 0$ and $u = 0$ when $y = 0$. (7)

- (b) Solve $(y^2 + x^2)y - xyq + zz = 0$. (7)

Module 2

- 13 (a) Derive one-dimensional wave equation. (7)

- (b) Solve the one dimensional heat equation $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ with boundary conditions $u(0, t) = 0^\circ\text{C}$, $u(1, t) = 0^\circ\text{C}$, and the temperature initially is $u(x, 0) = x(1 - x)$. (7)

- 14 (a) A homogeneous rod of conducting material of length l has its ends kept at zero temperature and the temperature initially is $u(x, 0) = f(x)$. Find the temperature $u(x, t)$ at any time. (7)

- (b) Solve the one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ subject to $u(0, t) = 0$, $u(1, t) = 0$, $u(x, 0) = 3 \sin(2\pi x)$, $\frac{\partial u}{\partial t}(x, 0) = 0$. (7)

Module 3

- 15 (a) Find all points where the Cauchy-Riemann equations are satisfied for the function $f(z) = xy$ and where it is analytic. Give reasoning for your answers. (7)

- (b) Show that $u = 2 + 3x - y + x^2 - y^2 - 4xy$ is harmonic. Also find its conjugate harmonic function v . (7)

- 16 (a) Show that the function $f(z) = \frac{1}{z}$ is analytic except at $z = 0$. (7)
- (b) Find the image of the triangular region bounded by $x = 1$, $y = 1$ and $x + y = 1$ under the transformation $w = z^2$. (7)

Module 4

- 17 (a) Evaluate $\int_C \frac{1}{z+4i(2z+1)} dz$, where C is $|z| = 2$ using Cauchy's integral formula. (7)

- (b) Find the Maclaurin series expansion of e^z using Cauchy's integral formula. (7)

- 18 (a) Evaluate $\int_C \frac{z^2 + 1}{(z-2)^2} dz$, where C is the circle $|z - 3| = 3$ using Cauchy's integral formula. (7)

- (b) Find the Taylor series expansion of $f(z) = \frac{1}{z}$ about $z = 2$. (7)

Module 5

- 19 (a) Obtain the Laurent series expansion of $\frac{1}{(z-2)(z-3)}$ in the region (i) $|z| < 1$ (ii) $2 < |z| < 3$ (iii) $|z| > 3$ (7)
- (b) Evaluate $\int_0^{\pi} \frac{1}{1+z^2} dz$. (7)
- 20 (a) Find the poles and residues of the function $\frac{1}{z^4-1}$. (7)
- (b) Evaluate $\int_0^{2\pi} \frac{1}{3+2\sin \theta} d\theta$. (7)